

Sales Forecasting & Business Analytics Report

Executive Summary

This report summarizes retail sales analysis, forecasting, anomaly detection, and product demand segmentation to support inventory and financial planning. Historical sales data was analyzed to identify trends, forecast future demand, detect unusual sales events, and group products by demand patterns. The results provide actionable recommendations for improving inventory management and business decision-making.

Key Findings

Technology generated the highest revenue, followed by Furniture and Office Supplies. Monthly sales showed seasonal peaks, especially during September–December. Sales generally increased year over year.

3-Month Forecast

Forecasts using SARIMA, Prophet, and XGBoost indicate continued positive sales trends with expected seasonal fluctuations. Forecast confidence intervals suggest moderate uncertainty but overall stable business growth.

Top Anomalies

Z-Score detection identified major sales spikes likely caused by festive promotions, large enterprise orders, or seasonal campaigns. Monitoring these events improves forecasting accuracy.

Demand Segmentation

K-Means clustering grouped products into demand segments such as High Demand, Stable Demand, Growing Demand, and Low Demand. High-demand products should receive priority inventory, while volatile products require cautious stock management.

Business Recommendations

1. Increase inventory for high-demand technology products before seasonal peaks.
2. Monitor anomaly periods to improve promotional planning.
3. Use demand clusters to optimize inventory and reduce holding costs.

Risk / Limitation

Forecast accuracy depends on historical trends and may be affected by unexpected market changes, economic conditions, or new customer behavior.